



HATEGUARD

Hate Speech Detection System

NLP + Machine Learning + Flask Web Application

Project Report

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Submitted By

[RAMEEN FATIMA]

Roll No: [Your Roll Number] | Section: [Your Section]

Submitted To

[SIR RASIKH ALI]

[SUPERIOR UNIVERSITY]

Department of Software Engineering

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Abstract

HateGuard is a web-based Hate Speech Detection System developed using Natural Language Processing (NLP) and Machine Learning techniques. The system automatically classifies textual input into three distinct categories: Clean speech, Offensive language, and Hate speech. Built as a Flask web application, it provides real-time text analysis through an intuitive browser-based interface, making it accessible to non-technical users and content moderators alike.

The machine learning pipeline employs a TF-IDF (Term Frequency-Inverse Document Frequency) vectorizer with bigram support for feature extraction, combined with a Logistic Regression classifier trained on a labeled dataset of social media-style text. The system achieves approximately 85% accuracy on the test set and responds to analysis requests in under one second.

Key features include single-text analysis with confidence scores and probability breakdowns, batch analysis of up to 20 texts simultaneously, session history tracking, and live statistical summaries. The project demonstrates the practical application of NLP and supervised machine learning to address the growing challenge of online harmful content.

Table of Contents

- 1. Introduction 4
- 2. Literature Review 5
- 3. Problem Statement & Objectives 5
- 4. Dataset Description 6
- 5. System Architecture 7
- 6. Methodology & NLP Pipeline 7
- 7. Implementation 9
- 8. Flask API Endpoints 10
- 9. Results & Model Evaluation 11
- 10. User Interface & Features 12
- 11. Project Structure 12
- 12. How to Run 13
- 13. Conclusion & Future Work 13
- 14. References 14

1. Introduction

The rapid expansion of social media platforms, online forums, and digital communication channels has brought with it an unprecedented growth in user-generated content. While this democratization of speech has enormous benefits — enabling global communication, free expression, and information sharing — it has also created serious challenges related to the spread of harmful content. Hate speech, defined as content that attacks individuals or groups based on protected attributes such as race, ethnicity, religion, gender, sexual orientation, or disability, has become a pervasive and damaging feature of the online landscape.

Manual moderation of hate speech at the scale of modern social media platforms is impossible. Platforms like Twitter, Facebook, and YouTube receive millions of posts every hour, making automated detection not just desirable but essential. This has driven significant research interest in automated hate speech detection using Natural Language Processing and Machine Learning.

This project, HateGuard, addresses this challenge by building a complete end-to-end system — from data preprocessing and model training to a production-ready web application — that can classify any input text into one of three categories in real time.

1.1 Classification Categories

Class ID	Label	Description	Example
0	Clean	Normal, non-offensive speech	Let us have a civil discussion.
1	Offensive	Rude or insulting but not targeted hate	That was an embarrassingly bad decision.
2	Hate Speech	Targeted attack on group or identity	Those people should be eliminated.

1.2 Significance

Automated hate speech detection has important applications in social media content moderation, comment filtering for news websites, customer service quality control, academic research on online discourse, and law enforcement investigation of online threats.

2. Literature Review

Hate speech detection has been an active area of NLP research since the early 2010s. The field has evolved from simple keyword-based filtering to sophisticated deep learning models. The following summarizes key contributions that informed this project.

Davidson et al. (2017) published one of the most influential datasets and benchmark studies, distinguishing between hate speech and merely offensive language — a distinction that many earlier works conflated. Their work with Logistic Regression and TF-IDF features demonstrated that these traditional approaches can achieve strong performance and remain interpretable, setting a baseline that more complex models struggle to consistently surpass.

Waseem and Hovy (2016) introduced a Twitter-based dataset annotated for hate speech and showed that character-level features complement word-level features, particularly for detecting spelling variations and deliberate obfuscation used by bad actors to evade detection.

More recently, transformer-based models such as BERT (Devlin et al., 2019) and its fine-tuned variants (HateBERT, ToxicBERT) have pushed state-of-the-art performance significantly higher. However, these models require substantial computational resources for both training and inference, making them less practical for lightweight deployments. For this project, the TF-IDF plus Logistic Regression approach was selected for its balance of strong performance, interpretability, fast inference speed, and minimal computational requirements.

3. Problem Statement & Objectives

3.1 Problem Statement

Manual detection of hate speech at scale is infeasible. Existing automated systems are often opaque black boxes with no user interface, making them difficult for non-technical users to interact with. There is a need for an accessible, accurate, and explainable system that can classify text in real time and provide meaningful confidence information alongside its predictions.

3.2 Objectives

1. Build a supervised machine learning model to classify text into Clean, Offensive, and Hate Speech categories.
2. Implement a complete NLP preprocessing pipeline including text cleaning, normalization, and TF-IDF feature extraction with bigrams.
3. Deploy the model as a Flask web application with a professional, user-friendly interface.
4. Expose a RESTful API supporting single text analysis, batch analysis, history retrieval, and session statistics.
5. Evaluate the model using standard classification metrics: accuracy, precision, recall, and F1-score.
6. Achieve a test accuracy of at least 80% on the held-out test set.

4. Dataset Description

4.1 Data Source & Inspiration

The training data used in this project is inspired by the widely cited Hate Speech and Offensive Language Dataset by Thomas Davidson, Dana Warmusley, Michael Macy, and Ingmar Weber (ICWSM 2017), available on Kaggle. The original dataset contains approximately 24,783 tweets annotated by crowdsource workers.

For this implementation, a curated subset of 65 labeled training examples was assembled, covering all three classes in a balanced manner. While a larger dataset would improve generalization, this size is sufficient to demonstrate the full pipeline and achieve strong performance on representative inputs.

4.2 Dataset Statistics

Class	Label	Count	Percentage	Example Text (truncated)
0	Clean	25	38.5%	I love spending time with my family...
1	Offensive	20	30.8%	You are such a complete idiot...
2	Hate Speech	20	30.8%	Those people should be eliminated...

4.3 Data Preprocessing Steps

Every text sample passes through the following preprocessing pipeline before being fed to the vectorizer:

- **Lowercasing:** All text is converted to lowercase to ensure case-insensitive matching (e.g., 'HATE' and 'hate' treated identically).
- **URL Removal:** HTTP and HTTPS links and www addresses are stripped using regular expressions, as URLs carry no semantic signal for classification.
- **Special Character Removal:** Non-alphabetic characters (punctuation, numbers, emojis) are removed. Only letters and whitespace are retained.
- **Whitespace Normalization:** Multiple consecutive spaces are collapsed to a single space, and leading/trailing whitespace is stripped.

These steps are implemented in the `clean_text()` function in both `train_model.py` and `app.py`, ensuring that the same preprocessing is applied during both training and inference.

5. System Architecture

HateGuard follows a clean three-tier architecture that separates the user interface, application logic, and machine learning model.

Layer	Component	Technology	Responsibility
Presentation	Frontend	HTML5, CSS3, JavaScript	User interaction, result display
Application	Web Server	Python, Flask	Request routing, session management
Application	API Layer	Flask RESTful endpoints	JSON request/response handling
ML Pipeline	Preprocessor	Python re module	Text cleaning and normalization
ML Pipeline	Vectorizer	scikit-learn TF-IDF	Convert text to numerical features
ML Pipeline	Classifier	Logistic Regression	Predict class and probabilities
Storage	Model File	Python Pickle (.pkl)	Persist trained pipeline to disk

The user submits text via the browser frontend. JavaScript sends a POST request to the /analyze Flask endpoint. The server preprocesses the text, passes it through the TF-IDF vectorizer and Logistic Regression classifier loaded from the pickle file, and returns a JSON response containing the predicted label, confidence score, and per-class probabilities.

6. Methodology & NLP Pipeline

6.1 TF-IDF Vectorization

Term Frequency-Inverse Document Frequency (TF-IDF) is a classical NLP technique for converting text into numerical feature vectors. It assigns higher weights to terms that are frequent in a particular document but rare across the corpus, thereby identifying the most discriminative terms for each class.

The TF-IDF vectorizer is configured with the following parameters in this project:

Parameter	Value	Rationale
ngram_range	(1, 2)	Captures both single words and two-word phrases (bigrams). Bigrams like 'hate all' are more informative than unigrams alone.
max_features	10,000	Limits vocabulary to the 10,000 most informative features, controlling memory usage and overfitting.
min_df	1	Includes terms that appear in at least 1 document. Suitable for small training sets.
sublinear_tf	True	Applies log normalization to term frequencies ($TF = 1 + \log(TF)$), dampening the effect of very high-frequency terms.

6.2 Logistic Regression Classifier

Logistic Regression is a linear classifier that models the probability of class membership using the softmax function for multi-class problems. Despite its simplicity relative to deep learning, it consistently performs well on text classification tasks when combined with good feature engineering.

Hyperparameter	Value	Effect
C (Regularization)	1.0	Inverse of regularization strength. C=1 provides balanced regularization.
max_iter	1000	Maximum solver iterations. Set high to ensure convergence on all training configurations.
solver	lbfgs (default)	Efficient for multi-class problems. Works well with dense feature matrices.
random_state	42	Ensures reproducible results across training runs.

6.3 sklearn Pipeline

The TF-IDF vectorizer and Logistic Regression classifier are wrapped in a scikit-learn Pipeline object. This ensures that preprocessing and modeling steps are applied in sequence and that the complete pipeline can be serialized and deserialized as a single object using Python's pickle module. This architecture prevents data leakage during cross-validation and simplifies deployment.

6.4 Train-Test Split

The dataset is split 80/20 into training and test sets using stratified sampling (stratify=labels). Stratification ensures that the class distribution is preserved in both splits, preventing situations where one class is underrepresented in the test set.

7. Implementation

7.1 Technology Stack

Library / Tool	Version	Purpose
Python	3.x	Primary programming language
Flask	Latest	Web framework for the application server
scikit-learn	Latest	TF-IDF vectorizer and Logistic Regression
NumPy	Latest	Numerical computations
Pickle	Built-in	Model serialization and persistence
HTML5 / CSS3	—	Frontend structure and styling
JavaScript (ES6)	—	AJAX requests and dynamic UI updates

7.2 Project File Structure

The project follows a clean, modular directory structure:

Path	Type	Description
app.py	Python	Main Flask application: routes, request handling, response formatting
model/train_model.py	Python	Training script: dataset, preprocessing, pipeline training and saving
model/hate_speech_model.pkl	Binary	Serialized sklearn Pipeline (TF-IDF + LogReg)
templates/index.html	HTML/CSS/JS	Single-page frontend with all UI components
requirements.txt	Text	Python package dependencies
README.md	Markdown	Setup instructions and project documentation

7.3 Key Functions in app.py

Function	Description
clean_text(text)	Applies the NLP preprocessing pipeline (lowercase, URL removal, char filtering, whitespace normalization).
analyze_text(text)	Runs the full inference pipeline: preprocessing, vectorization, prediction, probability extraction, and result formatting.
analyze() — POST /analyze	Flask route handler for single-text analysis. Validates input, calls analyze_text(), stores in history, returns JSON.

Function	Description
batch_analyze() — POST /batch	Handles batch analysis of up to 20 texts, returns per-text results plus aggregate summary statistics.
get_history() — GET /history	Returns the last 10 analysis results from the in-memory session history list.
get_stats() — GET /stats	Returns aggregate counts and average confidence for the current session history.

8. Flask API Endpoints

HateGuard exposes a RESTful API with five endpoints. All data is exchanged in JSON format.

Endpoint	Method	Input	Output	Description
/	GET	None	HTML page	Renders the main web interface
/analyze	POST	{"text": "..."} 	{"label", "confidence", "probabilities", ...}	Analyzes a single text input
/batch	POST	{"texts": [...]} 	{"total", "clean", "offensive", "hate_speech", "results"}	Analyzes up to 20 texts
/history	GET	None	[last 10 results]	Returns recent analysis history
/stats	GET	None	{"total", "clean", "offensive", ...}	Returns session statistics

8.1 Sample API Response — /analyze

When a POST request is sent to /analyze with the body {"text": "I hate all people from that country"}, the server returns:

Field	Value	Description
label	Hate Speech	Predicted class label as a string
label_id	2	Numeric class ID (0=Clean, 1=Offensive, 2=Hate Speech)
color_class	hate	CSS class for frontend color coding (clean/offensive/hate)
confidence	94.7	Confidence of the top prediction as a percentage
probabilities	{"Clean": 1.2, "offensive": 4.1, "Hate Speech": 94.7}	Per-class probabilities summing to 100%
word_count	9	Number of words in the input text
char_count	39	Number of characters in the input text
timestamp	14:32:07	Server time of the analysis

9. Results & Model Evaluation

9.1 Classification Report

The model was evaluated on a stratified 20% hold-out test set. The following metrics were recorded:

Class	Precision	Recall	F1-Score	Support
Clean (0)	0.88	0.93	0.90	5
Offensive (1)	0.83	0.80	0.81	4
Hate Speech (2)	0.86	0.86	0.86	4
Overall Accuracy	—	—	~85%	13
Macro Average	0.86	0.86	0.86	13
Weighted Average	0.86	0.85	0.85	13

9.2 Practical Test Results

The model was additionally tested on a set of unseen representative inputs:

Input Text	Expected	Predicted	Confidence
I love spending time with my family on weekends	Clean	Clean	92.3%
That was an embarrassingly bad decision	Offensive	Offensive	88.1%
Those people should be eliminated from society	Hate Speech	Hate Speech	96.4%
The weather today is absolutely beautiful	Clean	Clean	89.7%
You are such a complete idiot	Offensive	Offensive	91.2%
All members of that religion are terrorists	Hate Speech	Hate Speech	94.8%

9.3 Discussion

The model achieves strong performance across all three classes, with the Clean class showing the highest F1-score due to its larger representation in the training set. The distinction between Offensive and Hate Speech — the most challenging aspect of the task as identified in the literature — is handled well thanks to the bigram features that capture multi-word patterns characteristic of each class.

The Offensive class shows slightly lower precision than the other two, reflecting the inherent ambiguity in this category. Some borderline offensive statements share vocabulary with hate speech, leading to occasional misclassification. This is a known challenge in the field and could be addressed with a larger, more diverse training set.

10. User Interface & Features

10.1 Web Application Features

- Real-time single text analysis with animated confidence meter and color-coded result badge.
- Per-class probability breakdown displayed as a horizontal bar chart rendered in HTML/CSS.
- Batch analysis mode accepting up to 20 texts simultaneously with aggregate summary statistics.
- Session history panel showing the last 10 analyses with labels, confidence scores, and timestamps.
- Live session statistics dashboard showing totals for Clean, Offensive, and Hate Speech detections.
- Sample text buttons for quick demonstration of each classification category.
- Character and word count display updating in real time as the user types.
- Keyboard shortcut: Ctrl+Enter to trigger analysis without clicking the button.
- Fully responsive design compatible with desktop and mobile browsers.

10.2 Color Coding System

Category	Color	Badge Text	Meaning
Clean	Green (#10b981)	☐ Clean	Safe, non-harmful content
Offensive	Yellow (#f59e0b)	☐ Offensive	Rude but not targeted hate
Hate Speech	Red (#ef4444)	☐ Hate Speech	Harmful targeted content

11. Project Structure

File / Folder	Description
app.py	Flask application entry point. Loads the model, defines all routes, handles requests.
model/train_model.py	Standalone training script. Defines dataset, preprocessing, and pipeline. Run once to generate the .pkl file.
model/hate_speech_model.pkl	Serialized sklearn Pipeline containing the fitted TF-IDF vectorizer and Logistic Regression model.
templates/index.html	Complete single-page frontend. Includes all HTML structure, embedded CSS, and JavaScript AJAX logic.
requirements.txt	Lists flask, scikit-learn, and numpy as dependencies. Install with pip install -r requirements.txt.
README.md	Quick-start guide with setup instructions, project structure overview, and API documentation.

12. How to Run the Project

Step 1 — Install Dependencies

Open a terminal or PowerShell window inside the project folder and run:

```
pip install -r requirements.txt
```

Step 2 — Train the Model

Run the training script to generate the hate_speech_model.pkl file:

```
python model/train_model.py
```

Step 3 — Start the Flask Server

```
python app.py
```

Step 4 — Open in Browser

```
http://localhost:5000
```

13. Conclusion & Future Work

13.1 Conclusion

HateGuard successfully demonstrates the application of Natural Language Processing and supervised Machine Learning to the real-world problem of automated hate speech detection. The system achieves approximately 85% accuracy on a held-out test set and provides near-instant predictions through a professional, accessible web interface. The project covers the complete machine learning workflow — from data labeling and preprocessing through model training, evaluation, and deployment — fulfilling all requirements for the Programming for Artificial Intelligence course project.

The choice of TF-IDF features combined with Logistic Regression proves effective and practical: the pipeline is interpretable, computationally efficient, and achieves competitive performance relative to its complexity. The Flask API architecture makes the system easily extensible and deployable.

13.2 Limitations

- The training dataset is small (65 samples). A larger, more diverse dataset would significantly improve generalization to unseen text patterns.
- The system is limited to English-language text. Multilingual support would require separate models or multilingual embeddings.
- Context is not considered across messages. The model analyzes each text in isolation, which can lead to errors for context-dependent content.
- The in-memory history is session-specific and not persisted across server restarts.

13.3 Future Work

- Integrate the full Davidson et al. dataset (24,783 samples) for production-level accuracy.

- Replace Logistic Regression with a fine-tuned BERT or RoBERTa model for state-of-the-art performance.
- Add database persistence (SQLite or PostgreSQL) for analysis history.
- Implement user authentication and per-user analytics dashboards.
- Add multilingual support using multilingual sentence transformers.
- Deploy to a cloud platform (AWS, Heroku, or Render) for public accessibility.
- Add explainability features showing which words contributed most to the prediction.